

Table A1: Rubric for Scoring Criteria (0 = Very Poor, 3 = Excellent)

Criterion	Score	Description
Detection Performance	0	Fails to identify anomalies; metrics (precision and recall) not reported.
	1	Basic detection with high false positives/negatives; metrics reported but not used for validation.
	2	Reasonable accuracy; metrics reported and used for validation but lacking thorough analysis.
	3	High accuracy with robust validation; metrics explicitly reported, thoroughly analysed, and central to evaluation.
Theoretical Foundations	0	No theoretical basis or justification.
	1	Weak grounding; methods poorly justified.
	2	Reasonably grounded in established theory; some justification provided.
	3	Strong theoretical basis; well-justified and supported by prior research.
Computational Efficiency	0	Not feasible for practical use; no efficiency considerations.
	1	Significant resource requirements; unsuitable for large-scale or real-time use.
	2	Acceptable efficiency; suitable for many applications.
	3	Highly efficient; fast, scalable, and suitable for real-time processing.
Interpretability	0	Completely opaque; black box with no explanation of decisions.
	1	Limited; some explanation provided but difficult to trust fully.
	2	Moderate; provides interpretable insights that aid understanding.
	3	Highly interpretable; transparent with clear visualisations and explanations.
Ecological Relevance	0	No practical ecological application.
	1	Addresses some ecological aspects but lacks direct applicability.
	2	Applicable to ecological questions with some validation.
	3	Directly addresses ecological problems with clear practical implications.